



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

31-05-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 31-05-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-1	Q.	1	2	3							
	A.	B	B	B							
SEC-2	Q.	1	2	3	4	5					
	A.	A,B,D	A,C,D	A,B,C,D	B,C,D	A,B,C					
SEC-3	Q.	1	2								
	A.	A	B								
SEC-4	Q.	1	2	3							
	A.	2200.00	3.00	9.37 to 9.38							
SEC-5	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8	1	9	6	4	3	6	5	5	0

CHEMISTRY

SEC-1	Q.	1	2	3							
	A.	C	D	B							
SEC-2	Q.	1	2	3	4	5					
	A.	A,C,D	A,D	A,B,C	A,D	A,B,D					
SEC-3	Q.	1	2								
	A.	C	D								
SEC-4	Q.	1	2	3							
	A.	252.65	1299.00	123.00							
SEC-5	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4	4	2	9	2	3	1	8	4	2

MATHEMATICS

SEC-1	Q.	1	2	3							
	A.	C	A	D							
SEC-2	Q.	1	2	3	4	5					
	A.	A,B	A,B,C,D	A,B	A,B,C	A,C					
SEC-3	Q.	1	2								
	A.	B	A								
SEC-4	Q.	1	2	3							
	A.	0000.25	0000.20	0003.50							
SEC-5	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	1	4	1	6	7	1	1	1	8

HINT – SHEET

PHYSICS

SEC-1

1. **Ans (B)**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-20} = \frac{1}{-40} \Rightarrow v = +40 \text{ mm}$$

$$m = \frac{-v}{u} = -\left(\frac{+40}{-20}\right) = +2$$

$$\Rightarrow \text{separation between the two images formed} = \sqrt{(20)^2 + (1)^2} \approx 20.025 \text{ mm}$$

2. **Ans (B)**

$$\sin \theta_C = \frac{\mu_2}{\mu_1}; 1 - \cos^2 \theta_C = \left(\frac{\mu_2}{\mu_1}\right)^2$$

$$1 - (\hat{n} \cdot \hat{p})^2 = \left(\frac{\mu_2}{\mu_1}\right)^2$$

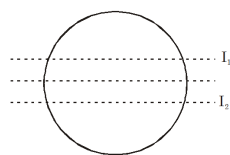
$$1 - \left[\frac{4}{\sqrt{25}}\right]^2 = \left(\frac{\mu_2}{\mu_1}\right)^2 \Rightarrow \frac{\mu_2}{\mu_1} = \frac{3}{5}$$

$$\Rightarrow \mu_2 = \frac{3\sqrt{3}}{5}$$

3. **Ans (B)**

$$\text{Let } \rho \frac{V}{2} = m$$

$$r_{CM} = \frac{m \left(R + \frac{3R}{8}\right) + 2m \left(R - \frac{3R}{8}\right)}{3m} = \frac{7R}{8}$$



Moment of inertia of sphere about a horizontal are passing through the hinge.

$$I = \left[I_1 + m \left(R + \frac{3R}{8}\right)^2 \right] + \left[I_2 + 2m \left(R - \frac{3R}{8}\right)^2 \right]$$

$$I = \left[\frac{2}{5} m R^2 - m \left(\frac{3R}{8}\right)^2 + m \left(\frac{11R}{8}\right)^2 \right] +$$

$$\left[\frac{2}{5} (2m) R^2 - 2m \left(\frac{3R}{8}\right)^2 + 2m \left(\frac{5R}{8}\right)^2 \right] = \frac{69mR^2}{20}$$

$$\tau = 6mgR\theta - 3mg \times \frac{7R}{8} \theta = \frac{27}{8} mgR\theta$$

$$I \frac{d^2\theta}{dt^2} + \frac{27}{8} mgR\theta = 0 \Rightarrow T = 2\pi \sqrt{\frac{46R}{45g}}$$

PHYSICS

SEC-2

1. **Ans (A,B,D)**

Impulse exist in action reaction pair during collision so magnitude will remain the same.

(a) Elastic collision principle

(b) Internal force

(d) If $m_1 \gg m_2$

Direction of velocity of motion does not change

$$\vec{v}_1 = (1 + e) \vec{v}_C = e \vec{v}_1 \quad (\vec{v}_C \approx \vec{v}_1)$$

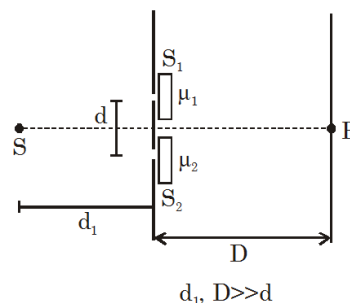
$$\vec{v}_1 = \vec{v}_C$$

2. **Ans (A,C,D)**

Basic knowledge to write angular momentum and kinetic energy of the system.

3. **Ans (A,B,C,D)**

Optical path difference due to



$$S_1 \text{ is } t(\mu_1 - 1) = 9\lambda \left(\frac{4}{3} - 1\right) = 3\lambda$$

Optical path differ due to S_2 is

$$= t(\mu_2 - 1) = 2\lambda \left(\frac{3}{2} - 1\right) = \lambda$$

Path diff. at point P = $3\lambda - \lambda = 2\lambda$

$$\text{Phase diff. } \frac{2\pi}{\lambda} \times 2\lambda = 4\pi$$

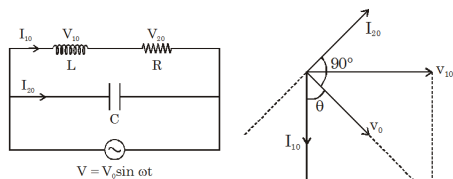
So net intensity at point P is

$$I_{\text{net}} = I_0 + I_0 + 2I_0 \cos 4\pi$$

$$I_{\text{net}} = 4I_0$$

Optical path is covered by S_1 is more so fringe will shift at upper side.

4. Ans (B,C,D)



For resonance

$$I_{20} = I_{10} \sin \theta$$

$$\frac{V_0}{X_C} = \frac{V_0}{\sqrt{X_L^2 + R^2}} \times \frac{X_L}{\sqrt{X_L^2 + R^2}}$$

$$\Rightarrow X_L^2 + R^2 = X_C X_L \Rightarrow L^2 \omega^2 + R^2 = \frac{L}{C}$$

$$\Rightarrow \omega^2 = \frac{1}{LC} - \left(\frac{R}{L}\right)^2$$

$$\Rightarrow \omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{L}\right)^2}$$

5. Ans (A,B,C)

By energy conservation

$$\frac{1}{2} \frac{Mm}{m+M} v^2 = \frac{1}{2} kx^2$$

$$\Rightarrow x = J \sqrt{\frac{m}{M(M+m)K}} = 5 \sqrt{\frac{1}{2 \times 3 \times 6}} = \frac{5}{6} \text{ m}$$

PHYSICS

SEC-4

1. Ans (2200.00)

$\Delta \ell_T = \alpha \ell \Delta T$ Extension ... (i)

$$\frac{mg}{S} = \frac{F}{S} = Y \frac{\Delta \ell_e}{\ell}$$

$$\Rightarrow \Delta \ell_e = \frac{mg\ell}{SY} \Rightarrow \text{compression ... (ii)}$$

If there is no change in length, it means extension due to temperature raise must be equal to the elastic compression due to weight.

So

$$\alpha \ell \Delta T = \frac{mg\ell}{SY}$$

$$\Rightarrow m = \frac{\alpha SY \Delta T}{g}$$

$$= \frac{1.1 \times 10^{-5} \times 10 \times 10^{-4} \times 10 \times 2.0 \times 10^{11}}{10}$$

$$= 2.2 \times 10^3 \text{ kg}$$

2. Ans (3.00)

$$L_1 = \frac{\mu_0 N_1^2 A}{l}, L_2 = \frac{\mu_0 N_2^2 A \mu_r}{l}$$

Since $V = -L \frac{dI}{dt}$, So

$$\frac{V_1}{V_2} = \frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2 \frac{1}{\mu_r} = \left(\frac{600}{200}\right)^2 \times \frac{1}{3} = 3$$

3. Ans (9.37 to 9.38)

Initially $V_{800} = V'_{1000}$

$$\Rightarrow V' = 0.8 V \text{ (V' is volume inside water)}$$

$$\Rightarrow H' = 0.8 H = 0.8 \times 50 \text{ cm} = 40 \text{ cm}$$

If we displace the block by dy inside water then

$$dy = dx \left[1 - \frac{a}{A}\right]$$

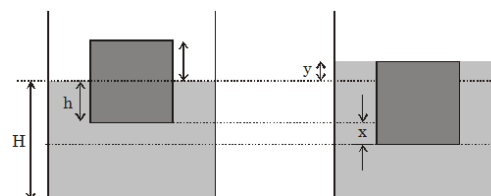
Where dx is the new dipped part of block on displacing the block by dy

$$\Rightarrow dW = \int_{0.4}^{0.5} [x(0.5)^2 1000g - (0.5)^3 800g] dx \frac{3}{4}$$

$$\Rightarrow W = 9.375 \text{ J}$$

Second Method :

a = Area of cube $A \Rightarrow$ Base area of container.



$$AH - ah = A(H + y) - a(h + x + y)$$

$$\Rightarrow y = \frac{a}{A-a} x \Rightarrow x + y = \frac{A}{A-a} x$$

$$F_{ex} = F_B - mg = \rho g a (h + x + y) - a^3 \rho_B g$$

$$= a(x + y) \rho g = a \left(\frac{A}{A-a}\right) x \rho g$$

$$W = \int_0^{x_0} F_{ex} dx = \frac{aA\rho g}{(A-a)} \times \frac{x_0^2}{2}$$

$$\text{Here } x_0 = \frac{0.2\sqrt{a}(A-a)}{A}$$

$$\Rightarrow W = \frac{aA\rho g}{(A-a)} \times \frac{a^2}{2} \times \frac{0.04 \times a(A-a)^2}{A^2}$$

$$= \frac{0.04a^2(A-a)}{2A} \rho g = \frac{0.04 \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{4} \times 1000 \times 10}{2 \times 1}$$

$$\frac{300}{32} = \frac{75}{8} = 9.375 \text{ Joule}$$

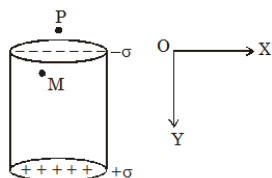
PHYSICS

SEC-5

1. Ans (8)

Assume that on circular face of cylinder, a charge q of surface charge density σ (in magnitude) appears. So,

$$Q = s\sigma, \quad \vec{E}_P = \frac{\sigma}{\epsilon_0} \hat{j}, \quad \vec{E}_{M(Q^-)} + \vec{E}_{M(Q)} + \vec{E}_{M(Q')} = 0 \dots (1)$$



$$\vec{E}_{P(q)} = \frac{q}{4\pi\epsilon_0 a^2} (\hat{j})$$

$$\vec{E}_{P(Q^-)} + \vec{E}_{P(q)} + \vec{E}_{P(Q^+)} = \frac{\sigma}{\epsilon_0} \hat{j} \dots (2)$$

From equation (1) and (2)

$$\vec{E}_{P(q)} = \frac{\sigma}{\epsilon_0} \hat{j}$$

$$\Rightarrow \sigma = \frac{q}{4\pi a^2} \Rightarrow Q = \frac{qS}{4\pi a^2}$$

$$|\vec{F}_{\text{cylinder}}| = \frac{qQ}{4\pi\epsilon_0} \left[\frac{1}{a^2} - \frac{1}{(a+l)^2} \right]$$

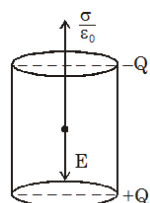
$$= \frac{qQ}{4\pi\epsilon_0 a^2 (a+l)^2} [(a+l)^2 - a^2]$$

$$= \frac{qQ (l^2 + 2al)}{4\pi\epsilon_0 a^2 (a+l)^2} \approx \frac{2qQal}{4\pi\epsilon_0 a^4}$$

$$= \frac{q\ell}{2\pi\epsilon_0 a^3} \times \frac{qS}{4\pi a^2} = \frac{q^2}{8\pi^2 \epsilon_0 a^5} V$$

Second Method :

Due to induction, $-Q = -s\sigma$ and $+Q = +s\sigma$ will appear at upper surface and lower surface of conductor. These two charged system will act like a capacitor for region inside the conductor, so due these charges, net electrostatic field at inside the conductor has magnitude $\frac{\sigma}{\epsilon_0}$. Since net field due to all charges inside the conductor is zero.



$$E = \frac{\sigma}{\epsilon_0} \Rightarrow \frac{q}{4\pi\epsilon_0 a^2} = \frac{\sigma}{\epsilon_0}; \sigma = \frac{q}{4\pi a^2}$$

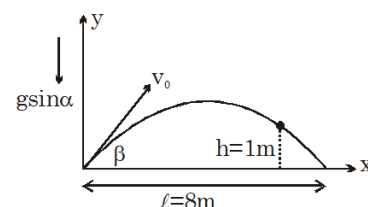
2. Ans (1)

$$\omega = \frac{\omega_0}{1 - a\theta/r_0}, \quad T = mr_0\omega_0\theta.$$

3. Ans (9)

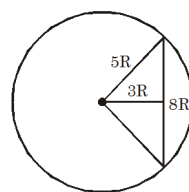
$$h = \ell \tan \beta - \frac{1}{2} \frac{g \sin^2 \alpha}{v_0^2 \cos^2 \beta}$$

$$v_0 = \sqrt{\frac{g \sin^2 \alpha}{2 \cos^2 \beta (\ell \tan \beta - h)}}$$



$$= \sqrt{\frac{10 \times \frac{4}{5} \times 8 \times 8}{2 \times \frac{4}{5} \times \frac{4}{5} (8 \times \frac{3}{4} - 1)}} = 9 \text{ m/s}$$

4. Ans (6)



$$\phi = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\rho \left(\frac{1}{2} 8R \cdot 3R \right) R}{\epsilon_0} = \frac{12\rho R^3}{\epsilon_0}$$

5. Ans (4)

The Half of the maximum current is equal to

$$i_1 = \frac{B\omega\ell^2}{4R} \text{ at } t = \frac{L}{R} \ln 2$$

$$\text{The torque at this instant} = \frac{B^2\omega\ell^4}{8R} = 4 \text{ N} \cdot \text{m}$$

6. Ans (3)

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{mgd}{I}}$$

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{(2m)(g) \left(\frac{3\ell}{4} \right)}{\left(\frac{m\ell}{3} + m\ell^2 \right)}} = \frac{1}{2\pi} \sqrt{\frac{9g}{8\ell}}$$

When disc is removed

$$f = \frac{1}{2\pi} \sqrt{\frac{mg \left(\frac{\ell}{2} \right)}{\frac{m\ell}{8}}} = \frac{1}{2\pi} \sqrt{\frac{3g}{2\ell}}$$

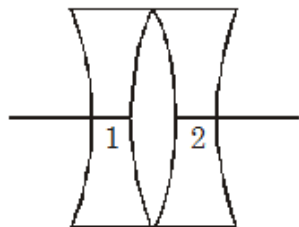
$$\frac{f}{f_0} = \sqrt{\frac{3g}{2\ell}} \sqrt{\frac{8\ell}{9g}} = \sqrt{\frac{4}{3}}$$

7. Ans (6)

$$\omega = \frac{1}{\sqrt{LC_{eq}}} = 100\pi$$

$$\Rightarrow T = 1/50 \text{ sec}$$

8. Ans (5)



9. Ans (5)

$$F = u \frac{dm}{dt} \text{ and } P = Fv$$

10. Ans (0)

$$i = \frac{\epsilon_0 a V}{d} (k-1) A \omega \sin \omega t$$

$$q = \int |i| dt$$

CHEMISTRY

SEC-1

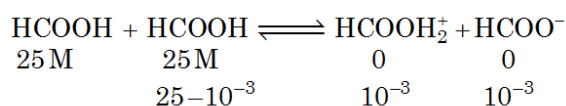
1. Ans (C)

B	>	Tl	>	Ga	>	Al	>	In
kJ/mole: 800		589		579		577		558

The last element (Tl) has exceptionally high first ionisation energy due to poor screening caused by f-electrons.

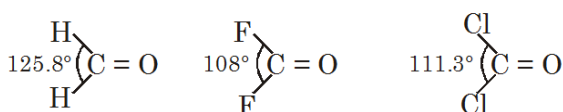
The second element (Al) has exceptionally low first ionisation energy due to good screening caused by s and p electrons.

2. Ans (D)



$$\% = \frac{10^{-3}}{25} \times 100 = 0.004\%$$

3. Ans (B)

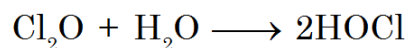
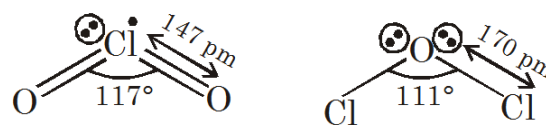


Higher the electronegativity of bonded atom, lesser is the bond angle between the bonds.

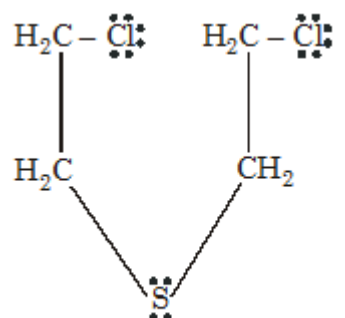
CHEMISTRY

SEC-2

1. Ans (A,C,D)

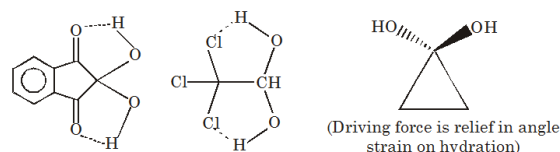


2. Ans (A,D)

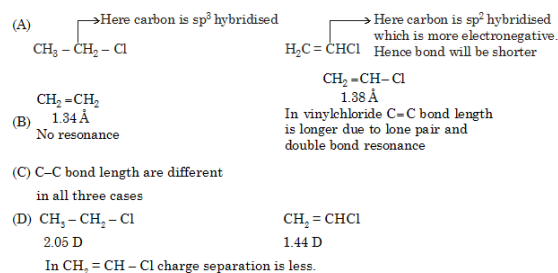


Mustard gas structure

3. Ans (A,B,C)



5. Ans (A,B,D)



CHEMISTRY

SEC-3

1. Ans (C)

E.W.

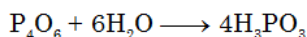
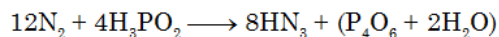
$\text{N}_2 = 42$

$\text{H}_3\text{PO}_2 = 33$

$\text{HN}_3 = 43$

$\text{P}_4\text{O}_6 = 27.5$

2. Ans (D)



6 moles H_2O reacts with 1 mole P_4O_6

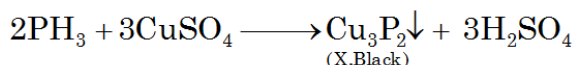
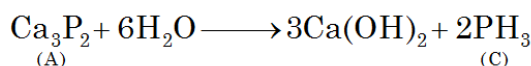
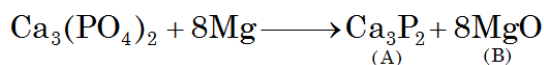
2 moles H_2O reacts with $\frac{1 \times 2}{6} = \frac{1}{3}$ mole P_4O_6

$\therefore$ Fraction of P_4O_6 left unreacted = $\frac{2}{3}$

CHEMISTRY

SEC-4

1. Ans (252.65)



3. Ans (123.00)

$\rightarrow V = \left(\frac{2\pi K e^2}{h} \right) \times \frac{Z}{n}$, so atomic number (Z) $\propto$ slope of the graph

So, statement (1) is true

$\rightarrow \lambda = \frac{h}{\sqrt{2qVm}}$, slope of the graph $\propto \frac{1}{\sqrt{m}}$

So, statement (2) is true

$\rightarrow$ From, $PV = nRT$

$V = \left(\frac{nR}{P} \right) T$, so, slope of graph $\propto \frac{1}{P}$

So, statement (3) is also correct.

$\rightarrow P = (nRT) \frac{1}{V}$, so, slope $\propto T$

So, statement (4) is incorrect.

So, statement (4) is incorrect.

CHEMISTRY

SEC-5

1. Ans (4)

pH range of indicator = 3.4 – 4.6

$\therefore$ pOH range of indicator = 9.4 – 10.6

$\therefore$ pK_b of indicator = $\frac{9.4 + 10.6}{2} = 10$

$\text{InOH} \rightleftharpoons \text{In}^+ + \text{OH}^-$

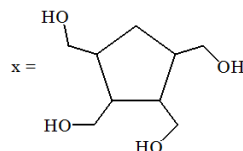
$\text{pOH} = \text{pK}_b + \log \frac{[\text{In}^+]}{[\text{InOH}]}$

For solution to appear in the colour of In^+ , $\text{pOH} \geq 10.6$

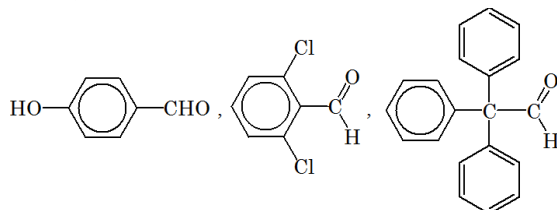
In limiting case, $10.6 = 10 + \log \frac{[\text{In}^+]}{[\text{InOH}]} \therefore$

$$\frac{[\text{In}^+]}{[\text{InOH}]} = 4$$

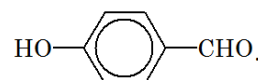
2. Ans (4)



3. Ans (2)

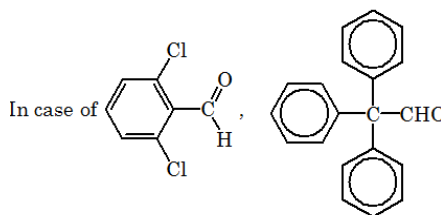


Do not contain any α -hydrogen but still fails to undergo cannizzaro reaction in case of

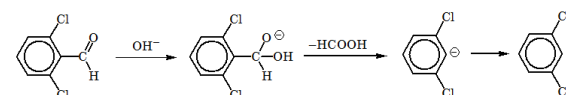


the corresponding hydroxide is strongly stabilized by – R effect of –CHO.

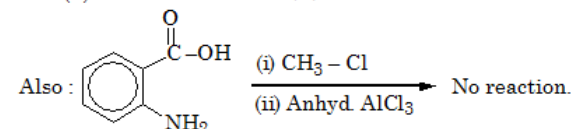
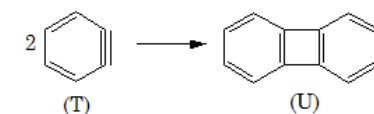
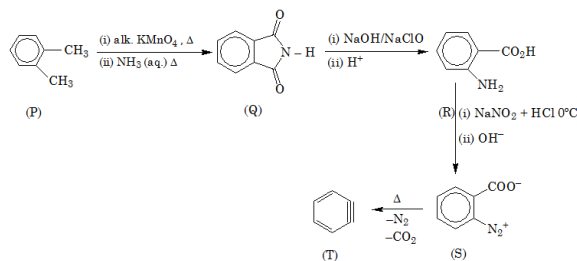
Consequently, the electrophilicity of the carbonyl carbon is reduced and the OH^- does not attack the carbonyl group.



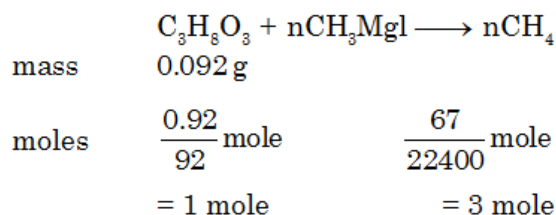
The intermediate formed due to attack of OH^- cleaves to give a stable anion.



4. Ans (9)



6. Ans (3)



So, on comparing each molecule of $\text{C}_3\text{H}_8\text{O}_3$ must give 3 molecule of CH_4

7. Ans (1)

$$\text{Rate} = k[\text{C}_3\text{H}_6^+] \quad (\text{i})$$

$$K_c (\text{for step 1}) = \frac{[\text{C}_3\text{H}_6][\text{C}_3\text{H}_6^*]}{[\text{C}_3\text{H}_6][\text{C}_3\text{H}_6]} = \frac{[\text{C}_3\text{H}_6^*]}{[\text{C}_3\text{H}_6]}$$

$$\therefore [\text{C}_3\text{H}_6^*] = K_c[\text{C}_3\text{H}_6]$$

Substituting in equation (i)

$$R = k \cdot K_c[\text{C}_3\text{H}_6]$$

$$R = k'[\text{C}_3\text{H}_6]$$

8. Ans (8)

$$v = \frac{V}{2\pi r}; V \propto \frac{Z}{n}; r \propto \frac{n^2}{Z}; v \propto \frac{Z^2}{n^3}$$

$$\frac{v_{\text{He}^+}}{v_{\text{H}}} = \frac{4}{27} \div \frac{1}{8} = \frac{32}{27} = \frac{2^5}{3^3}$$

$$x = 5$$

$$y = 3$$

$$x + y = 8$$

9. Ans (4)

b, c, d, f

10. Ans (2)

One mole of the above compound (X) on hydrolysis followed by heating produces 4 moles

of CO_2 .

Hence, 0.5 M of X produce 2 moles of CO_2 .

MATHEMATICS

SEC-1

1. Ans (C)

$$\frac{1}{3} \frac{d}{dx} \left\{ e^{x^3} (y^3 - 1) \right\} + \frac{1}{3} \frac{d}{dx} (e^{y^3}) = 0$$

2. Ans (A)

Each prime number can go in one of the five

slots - x_1, x_2, x_3, x_4 and none of these.

3. Ans (D)

A, B, C are angles of a triangle

$$\Rightarrow \sin 2A + \sin 2B + \sin 2C$$

$$= 4 \sin A \sin B \sin C$$

$$\text{Now, } 4^4 \left(\frac{\sin^{12} B + \sin^{12} C}{(\sin 2A + \sin 2B + \sin 2C)^4} \right)$$

$$= \frac{b^{12} + c^{12}}{(abc)^4}$$

$$= \frac{b^4 \left(\frac{b}{c} \right)^4 + c^4 \left(\frac{c}{b} \right)^4}{a^4} = \frac{\alpha^4 + \beta^4}{\left(\frac{\alpha + \beta}{2} \right)^4}$$

$$\left(\text{using } \frac{\alpha^4 + \beta^4}{2} \geq \left(\frac{\alpha + \beta}{2} \right)^4 \right)$$

$$\geq 2$$

$\Rightarrow$ Possible value = 3

MATHEMATICS

SEC-2

1. Ans (A, B)

Method 1 :

Any combination of 4 points gives 3 fresh point of intersection.

$$= {}^nC_4 \times 3 \Rightarrow (A)$$

Method 2 :

$$\text{Number of lines} = {}^nC_2 = m$$

number of fresh points of intersection

$$= {}^mC_2 - n \times {}^{n-1}C_2 \Rightarrow (B)$$

$$\frac{{}^nC_2 \cdot {}^{n-2}C_2}{2} = \frac{n(n-1)}{2} \times \frac{(n-2)(n-3)}{2 \times 2}$$

$$= \frac{n(n-1)(n-2)(n-3)}{8} = 3 \times {}^nC_4 \Rightarrow$$

(A)

2. Ans (A,B,C,D)

$$f(1-x) = f(1+x)$$

$$\Rightarrow f'(1-x) + f'(1+x) = 0$$

$$\text{put } x = 0 \Rightarrow f'(1) = 0$$

$$\text{Given } f'\left(\frac{1}{4}\right) = 0 \Rightarrow f'\left(\frac{7}{4}\right) = 0$$

$$f'\left(\frac{1}{2}\right) = 0 \Rightarrow f'\left(\frac{3}{2}\right) = 0$$

Applying Rolle's theorem in

$$\left(\frac{1}{4}, \frac{1}{2}\right), \left(\frac{1}{2}, 1\right), \left(1, \frac{3}{2}\right), \left(\frac{3}{2}, \frac{7}{4}\right)$$

$f''(x) = 0$ has at least 4 roots in (0,2)

$\Rightarrow$ (A)

$$I = \int_{-2}^2 (x^5 + x^3) f(1+x) dx$$

$$I = \int_{-2}^2 ((-x)^5 + (-x)^3) f(1-x) dx$$

$$\Rightarrow I + I = \int_{-2}^2 f(1+x) \times 0 dx = 0$$

$$\Rightarrow I = 0 < \int_0^2 \frac{1}{1+2^{f(x)}} dx \quad (\text{B})$$

$$\text{Suppose } F(x) = \frac{f'(x)}{x}$$

$$F(1) = F\left(\frac{3}{2}\right) = 0$$

Applying Rolles theorem

$$F'(c) = 0 \text{ for at least one } c \text{ in } \left(1, \frac{3}{2}\right)$$

$$\Rightarrow c \cdot f''(c) - f'(c) = 0$$

$$\Rightarrow c \cdot f''(c) = f'(c)$$

$$\text{for at least one } c \text{ in } \left(1, \frac{3}{2}\right)$$

$$\text{Suppose } G(x) = x \cdot f(x)$$

$$G\left(\frac{3}{2}\right) = G\left(\frac{7}{4}\right) = 0$$

$$G'(x) = 1 \cdot f'(x) + x f''(x)$$

$$\text{Applying Rolle's theorem in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

$$\Rightarrow f'(c) + c f''(c) = 0 \text{ for at least one } c$$

$$\text{in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

3. Ans (A,B)

$$I = \int_0^{\pi/2} \frac{\cos \theta}{10 - 6 \cos \theta} d\theta < \frac{1}{4} \int_0^{\pi/2} \cos \theta d\theta < 1$$

$$\frac{\cos \theta}{10 - 6 \cos \theta} > \left| \frac{\cos(\pi - \theta)}{10 - 6 \cos(\pi - \theta)} \right|$$

$$\text{for } \theta \in \left[0, \frac{\pi}{2}\right) \Rightarrow I > 0$$

4. Ans (A,B,C)

For area of parallelogram PQRS to be least line L

makes angle $\frac{\pi}{4}$ with L_1 and L_2

5. Ans (A,C)

Centre of the circle is at (a,a)

$$a^2 + (a-2)^2 = (a-4)^2 + (a-3)^2$$

$$\Rightarrow a = \frac{21}{10}$$

MATHEMATICS

SEC-3

1. Ans (B)

$$D_5$$

2. Ans (A)

$${}^5C_1 \{ {}^4C_1 \cdot D_4 + {}^4C_1 \cdot ({}^3C_1 \cdot D_3 + {}^3C_1 \times {}^2C_1) \}$$

MATHEMATICS

SEC-4

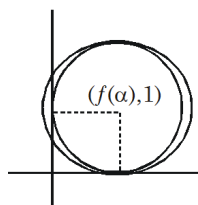
1. Ans (0000.25)

$$p = \frac{{}^5C_1}{{}^7C_3} \cdot \frac{2}{3} \cdot \left(\frac{1}{2}\right)^4$$

2. Ans (0000.20)

$$L = \text{circumference of unit circle} = 2\pi$$

3. Ans (0003.50)



From the figure it is clear that for $\alpha \geq \frac{\pi}{2}$ circle should either lie in second quadrant or touch the positive imaginary axis.

$$\Rightarrow f(\alpha) \leq 1$$

$$\alpha^2 - 7\alpha + 13 \leq 1$$

$$\alpha^2 - 7\alpha + 12 \leq 1$$

$$3 \leq \alpha \leq 4$$

MATHEMATICS

SEC-5

1. **Ans (3)**

$${}^3C_1 \times D_3 + {}^3C_1 \times D_2 \times {}^2C_1$$

2. **Ans (1)**

Put $x = r \cos \theta$; $y = r \sin \theta$

$$\Rightarrow r^2 = \frac{\sin 2\theta}{\cos^4 \theta + \sin^4 \theta}$$

$$= \frac{\sin 2\theta}{1 - \frac{1}{2}\sin^2 \theta}$$

$$= \frac{2 \sin 2\theta}{1 + \cos^2 2\theta}$$

$$\text{Required area} = 2 \int_0^{\pi/2} \frac{1}{2} r^2 d\theta$$

$$= \int_0^{\pi/2} \frac{2 \sin 2\theta}{1 + \cos^2 2\theta} d\theta = \frac{\pi}{2}$$

3. **Ans (4)**

Exactly four girls are together – Exactly four girls are together and $B_1 B_2$ are together

$$p = \frac{5! \times 8! \times 58}{13!}$$

4. **Ans (1)**

$$\tan \alpha = \frac{3}{5}, \tan \beta = \frac{5}{4 \cos \alpha} \frac{\sqrt{34}}{4}$$

5. **Ans (6)**

$$r_1 = 3r$$

$$\Rightarrow 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$= 3 \times 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\Rightarrow \tan \frac{B}{2} \cdot \tan \frac{C}{2} = \frac{1}{3}$$

$$\Rightarrow \tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{A}{2} \cdot \tan \frac{C}{2} = 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

$$\text{so, } 9 \left(\tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{A}{2} \cdot \tan \frac{C}{2} \right)$$

$$= 9 \times \frac{2}{3} = 6$$

6. **Ans (7)**

$$\Delta = \begin{vmatrix} \gamma^2 & \beta^2 & 2\beta\gamma - \alpha^2 \\ 2\alpha\gamma - \beta^2 & \alpha^2 & \gamma^2 \\ \alpha^2 & 2\alpha\beta - \gamma^2 & \beta^2 \end{vmatrix}$$

$$= (\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma)^2$$

7. **Ans (1)**

Denominator

$$= {}^nC_0 {}^{n+k-1}C_{n-1} + {}^nC_1 {}^{n+k-2}C_{n-1}$$

$$+ {}^nC_2 {}^{n+k-3}C_{n-1} + \dots + {}^nC_{k-1} {}^{n-1}C_{n-1}$$

= coefficient of x^{n-1} in

$${}^nC_0 (1+x)^{n+k-1} + {}^nC_1 (1+x)^{n+k-2}$$

$$+ {}^nC_2 (1+x)^{n+k-3} + \dots + {}^nC_{k-1} (1+x)^n$$

$$+ {}^nC_k (1+x)^{n-1}$$

= coefficient of x^{n-1} in

$${}^nC_0 (1+x)^{n+k-1} + {}^nC_1 (1+x)^{n+k-2}$$

$$+ {}^nC_2 (1+x)^{n+k-3} + \dots + {}^nC_{k-1} (1+x)^n$$

$$+ {}^nC_k (1+x)^{n-1}$$

$$+ {}^nC_{k+1} (1+x)^{n-2} + {}^nC_{k+2} (1+x)^{n-3} \dots$$

$$\dots + {}^nC_n (1+x)^{k-1}$$

= coefficient of x^{n-1} in

$$(1+x)^{n+k-1} (1 + (1+x)^{-1})^n = (1+x)^{k-1} (x+2)^n$$

= coefficient of x^{n-1} in

$$\sum_{r=1}^n {}^nC_r 2^r x^{n-r} (1+x)^{k-1} = \sum_{r=1}^n 2^r {}^nC_r x^{k-r} C_{r-1}$$

8. **Ans (1)**

Let r_1, r_2 and r are in-radius of $\triangle ABD$, $\triangle ADC$

and $\triangle ABC$ respectively

$$r_1 = \frac{c}{a} \times r; r_2 = \frac{b}{a} \times r \text{ and}$$

$$EF = \sqrt{2(r_1^2 + r_2^2)} = \sqrt{2}r = AI$$

9. Ans (1)

$$\begin{aligned}
 & \sum_{r=1}^m \frac{(m+1)(r-1)m^{r-1}}{r^m C_r} \\
 &= \sum_{r=1}^m \left(\frac{[mr - (m-r+1)] m^{r-1}}{r^m C_r} \right) \\
 &= \sum_{r=1}^m \left(\frac{m^r}{m C_r} - \frac{m^{r-1} (m-r+1) r! (m-r)!}{r \cdot m!} \right) \\
 &= \sum_{r=1}^m \left(\frac{m^r}{m C_r} - \frac{m^{r-1} (m-r+1)! (r-1)!}{m!} \right) \\
 &= \sum_{r=1}^m \left(\frac{m^r}{m C_r} - \frac{m^{r-1}}{m C_{r-1}} \right) = m^m - 1
 \end{aligned}$$

10. Ans (8)

Let the variable chord be $x \cos \alpha = p$. Let this chord intersect the hyperbola in A and B. the joint equation of OA and OB is

$$\begin{aligned}
 \frac{x^2}{4} - \frac{y^2}{8} &= \left(\frac{x \cos \alpha + y \sin \alpha}{p} \right)^2 \\
 \Rightarrow \left(\frac{1}{4} - \frac{\cos^2 \alpha}{p^2} \right) x^2 - \left(\frac{1}{8} + \frac{\sin^2 \alpha}{p^2} \right) y^2 \\
 &\quad - \frac{2 \sin \alpha \cos \alpha}{p^2} xy = 0 \\
 \Rightarrow \frac{1}{4} - \frac{\cos^2 \alpha}{p^2} - \frac{1}{8} - \frac{\sin^2 \alpha}{p^2} &= 0 \Rightarrow p^2 = 8
 \end{aligned}$$

.

The variable line touches the fixed circle, thus perpendicular distance of (0,0) = Radius

$\therefore$ Equation of the circle is $x^2 + y^2 = 8$